

Malik AlNajjar

Software Engineer | Full-Stack & Data

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SUMMARY

Software Engineer at MicroMobility in Jeddah, running the production booking, check-in and point-of-sale platform behind its group rides (PostgreSQL, Supabase, JavaScript, Cloudflare), and currently on a co-op in Sela's IT department. B.Sc. Software Engineering, First Class Honours (GPA 4.97/5.0); AWS-certified in data engineering and solutions architecture. Bilingual Arabic/English (IELTS 7.5).

EXPERIENCE

MicroMobility, Jeddah | live app: micromobilityrentals.pages.dev Jun 2026 – Present
Software Engineer

- Build and run the company's booking and operations platform: customer booking with Google and Apple sign-in, staff check-in, point of sale, inventory and refunds, used every ride night.
- Design the PostgreSQL backend on Supabase: row-level security on every table, staff-only functions, real-time sync across staff devices and 85+ versioned migrations.
- Deliver NFC bike check-in, Apple Wallet passes and a 10-language interface, including Arabic and Urdu right-to-left.
- Automate releases with CI/CD: GitHub Actions runs lint, type checks and 190+ Playwright end-to-end specs; Cloudflare deploys.

Sela, Saudi Arabia Jul 2026 – Present
IT Co-op

PROJECTS

AUJ, Smart Retail Shelf System | Senior Capstone 2025 – 2026

ESP32-S3, MQTT, FastAPI, PostgreSQL, LightGBM, YOLOv12n, WebSocket, React

- Built a dual-sensor IoT pipeline: a 10 Hz load-cell state machine on ESP32-S3, MQTT into PostgreSQL and 29 FastAPI endpoints, updating the dashboard within 5 seconds of a pick.
- Forecast per-SKU demand with LightGBM quantile models, repriced through a 5-rule audited engine within $\pm 15\%$ and verified every event with YOLOv12n; 143 pytest tests, zero PII.

Sadeed, Data Quality Platform | GASTAT Hackathon National Finalist 2026

FastAPI, Redis, Great Expectations, One-Class SVM, Qwen, Ollama | github.com/malikaalnajjar/Sadeed_GASat_Hackathon

- Designed a real-time 3-layer validator, shipped as a browser extension, for Arabic government survey data: Great Expectations checks, a One-Class SVM gate and a local LLM that explains each conflict.
- Raised anomaly detection from 23% to 87.5% (F1 0.829 vs 0.374) at over 97% precision, with no data leaving the government network.

Tayseer, Autonomous Smart Wheelchair | AVITHON (GACA), Top 15 of 500+ teams 2025

- Fused LiDAR, Intel RealSense and wheel-encoder data with ROS on Jetson Nano edge AI as data scientist in a team of five, backed by field research at King Abdulaziz International Airport.

Masra, Smart Adventure Tourism Platform | Aseer Tabtaker 2025

- Developed the AI layer in a team of five: AWS IoT Core early warning and scikit-learn models on SageMaker for a Flutter trail app.

SKILLS

Web & Backend: JavaScript, TypeScript, React, FastAPI, REST APIs, WebSocket, PostgreSQL (PL/pgSQL), Supabase, Cloudflare, PWA
Data & ML: Python (Pandas, NumPy, scikit-learn), SQL, Great Expectations, Redis, LightGBM, One-Class SVM, YOLO, Ollama, Power BI
Cloud, DevOps & Testing: AWS, Docker, CI/CD (GitHub Actions), Playwright, pytest, Git, Linux, Bash, Agile/Scrum
IoT & Edge: ESP32-S3, MQTT, Jetson Nano, ROS, LiDAR | **Also trained in:** Kafka, dbt, Iceberg, AWS Glue, Athena, Tableau, R

EDUCATION

University of Business and Technology, Jeddah 2026

Bachelor of Science in Software Engineering | First Class Honours | GPA 4.97/5.0

CERTIFICATIONS

AWS Certified Data Engineer – Associate (DEA-C01, Mar 2026) | AWS Certified Solutions Architect – Associate (SAA-C03, Feb 2026) | AWS Certified Cloud Practitioner (CLF-C02, Dec 2025) | DAMA CDMP (Jan 2026) | PMI CAPM (Jan 2026) | CompTIA Data+ (Nov 2025) | IBM Data Engineering Certificate (Dec 2025) | Google Data Analytics (Dec 2025) | Google Project Management (Jan 2026) | DataCamp Associate Data Engineer in SQL (Jun 2025) | McKinsey Forward Program (Jul 2025) | KAUST Advanced AI Specialization (Feb 2025)

AWARDS, LEADERSHIP & VOLUNTEERING

- Ranked Top 15 of 500+ teams at AVITHON 2025 (GACA), reached the GASTAT Hackathon national finals (2026) and placed 2nd of 30+ teams in the Project Management Championship (Beyond, Dec 2025).
- Led the GDG On Campus UBT outreach team (Oct 2025 – Jun 2026): 100+ students reached and 40%+ attendance growth.
- Managed a SAR 5,000 budget as ACM UBT Vice Chair and Treasurer (Feb 2025 – Dec 2025), delivering 3 to 5 events per cycle.
- Safeguard broadcast and intellectual-property rights at the AFC Champions League Elite Finals as an AFC Local Organising Committee volunteer (Apr 2026 – Present), after the AFC Asian Cup U-23 (Jan 2026).
- Volunteered at the Falling Walls Foundation (Sep 2025) and the APCG 2026 conference at UBT (Feb 2026).